Critic-Free Group-Relative Policy Optimization for Anti-Sway Control of a Double-Pendulum Crane

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GitHub: https://github.com/Zephyr-sue/rl_crane_antisway

Abstract—We formulate anti-sway transport with an overhead crane, where a trolley carries a hook and a payload (a double pendulum), as a continuous-control RL problem. Payload mass, cable lengths, initial sway and wind gusts are randomized. We adapt Group Relative Policy Optimization (GRPO) [1], originally proposed for LLM reasoning, to physical control. GRPO has no value network: G rollouts replay the same randomized scenario on a GPU-batched simulator and serve as each other's baseline. On 800 held-out test episodes in 8 conditions, including out-of-distribution plants and doubled gusts, GRPO succeeds in 94% of random scenarios, against 62% for the RBF neural-network adaptive controller of [2]. It also settles 1.7 s faster and wins in every condition.

I. Problem and Environment

Dynamics. The state is $q = [x, \theta_1, \theta_2]$: trolley position, hook angle and payload angle. The Euler–Lagrange equations $M(q)\ddot{q} = F(q, \dot{q}) + \tau$ are taken from [2], with viscous damping. They are integrated with RK4 at $\Delta t = 0.02\,\text{s}$ and are valid for large angles. A horizontal gust F_d on the payload enters through $\tau = J^\top F_d = F_d[1,\, L_1 \cos\theta_1,\, L_2 \cos\theta_2]^\top$. Tests check energy conservation without damping (drift $< 10^{-3}$) and that the energy change equals the gust's work.

Task. An episode lasts 12 s (600 steps). Its scenario is sampled at reset:

- payload $m_2 \in [0.5, 4]\,\text{kg}$ and cable lengths $L_1 \in [0.6, 1.4]\,\text{m}$, $L_2 \in [0.4, 1.2]\,\text{m}$;
- a target at a distance of $\pm[1, 4]\,\text{m}$;
- initial sway up to $\pm 30°$;
- up to three gusts of 2–6 N lasting 0.2–0.5 s, occurring during transport (the final 4 s are calm).

The gust schedule is pre-sampled, so the scenario is exactly replayable. End-stops at $|x| = 6\,\text{m}$ stop the trolley. During training a curriculum widens all ranges from 30% to 100% over the first 30% of training.

II. RL Formulation

State space $\mathcal{S} \subset \mathbb{R}^{13}$. The observation, with every entry normalized, is

$$s = [e, \dot{x}, \sin\theta_1, \cos\theta_1, \dot{\theta}_1, \sin\theta_2, \cos\theta_2, \dot{\theta}_2, e_p, u_{t-1}, m_2, L_1, L_2],$$

where:

- $e = x - x^*$ is the trolley's position error;
- e_p is the payload's horizontal position error;
- u_{t-1} is the previous action;
- (m_2, L_1, L_2) are the plant parameters, as measured by a load cell and a rope encoder. An ablation hides them (10-D).

Sin/cos encoding avoids angle wrap-around.

Action space $\mathcal{A} = [-1, 1]$ (continuous). The trolley force is $F = 30u\,\text{N}$.

Reward. With settled indicator σ_t ($|e| < 5\,\text{cm}$, $|\theta_{1,2}| < 2°$, $|\dot{x}| < 5\,\text{cm/s}$):

$$\begin{aligned} r_t = -\big[&|e| + 0.05\dot{x}^2 + 2(\theta_1^2 + \theta_2^2) + 0.1(\dot{\theta}_1^2 + \dot{\theta}_2^2) \\ &+ 0.01u^2 + 0.1(u_t - u_{t-1})^2\big] \\ &- 0.1(1 - \sigma_t) + \sigma_t - 10\,\mathbb{1}_{\text{end-stop}}. \end{aligned}$$

The linear $|e|$ term gives a constant pull toward the target, and together with the per-step time cost it rewards fast transport. The sway and rate terms penalize the coupled oscillations. The Δu term asks for smooth force. The settle bonus rewards finishing with the payload at rest.

III. Algorithms

GRPO for control (ours). Each iteration samples $N = 64$ scenarios ("prompts"). For each one it rolls out $G = 32$ trajectories ("outputs") of a Gaussian MLP policy (2×128, tanh), giving 2048 cranes simulated in parallel in PyTorch on the GPU. There is no critic. Advantages are group-relative, and we compare three variants:

- outcome [1]: $\hat{A}_i = (R_i - \bar{R})/\sigma_R$;
- process [1]: the suffix sum of group-normalized rewards;
- rtg (ours, default): the discounted return-to-go, normalized across the group at each timestep, $\hat{A}_{i,t} = (G_{i,t} - \overline{G_{\cdot,t}})/\sigma_{G_{\cdot,t}}$. This is a per-time baseline suited to dense rewards.

Because all members of a group share the scenario, the baseline removes the variance caused by the plant and the disturbances, which is the job a learned critic would do in PPO. The loss is the GRPO objective (clipped ratio, $\epsilon = 0.2$) with the unbiased KL estimator $\frac{\pi_{ref}}{\pi} - \log\frac{\pi_{ref}}{\pi} - 1$ ($\beta = 0.04$). The reference policy is refreshed every 10 iterations (iterative GRPO).

Baseline: RBF adaptive control [2]. This controller is not learned by RL; it serves as the comparison. Its control law is $\tau = M_0(q)(\ddot{q}_d - K_v\dot{e} - K_p e) + C_0\dot{q} + G_0 - \hat{f}$, built on the nominal model (m_2=2, L_1=1, L_2=0.8) and following a quintic reference.

TABLE I: GRPO vs RBF adaptive control (100 episodes per condition). t_s is averaged over settled episodes. OOD: m_2=5 kg, L_1+L_2=2.8 m.

Condition	Success (%)		t_s (s)		Max sway (°)	
	GRPO	RBF	GRPO	RBF	GRPO	RBF
Nominal	100	71	7.1	9.2	29.8	24.8
Random	94	62	7.1	8.8	29.7	26.6
Heavy	100	72	6.7	8.5	26.9	24.0
Light	48	23	9.0	9.3	50.2	48.5
Long cable	73	31	8.9	10.5	28.2	27.8
Short cable	100	75	6.4	8.6	37.3	27.1
Strong gusts	92	48	7.0	9.0	33.3	29.2
OOD	81	56	8.2	9.9	23.7	23.6

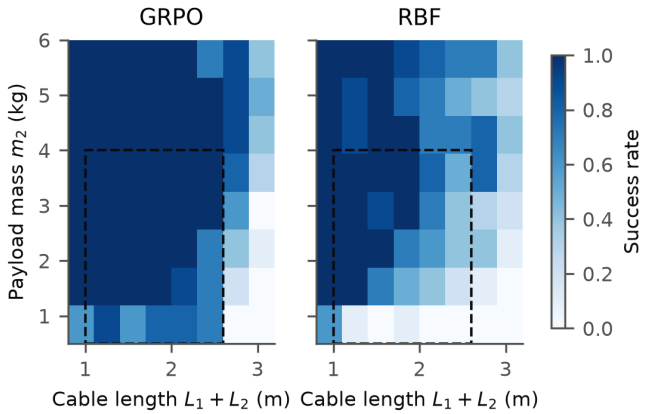


Fig. 1: Success rate over payload mass and cable length (10 episodes per cell). The dashed box is the training range.

- RBF compensator: $\hat{f} = W^\top h([e, \dot{e}])$ with 11 Gaussian units. The weights adapt online as $\dot{W} = \gamma\, h\, [e, \dot{e}]^\top P B$, where $PA + A^\top P = -I$.
- Underactuation: only the trolley row of τ is actuated.
- Gains: with the paper's positive sway gains the sway was not damped (12% success). A grid search gave K_p=diag(5, −22, −22), K_v=diag(6, 0.5, 0.5), γ=5.

IV. Results

Protocol. Each condition uses 100 seeded test scenarios that are never used in training. An episode succeeds if the crane is settled during the final second; settling time t_s is the first instant after which it stays settled. Both controllers run on the same scenarios and gust schedules.

GRPO vs RBF. GRPO succeeds more often in every condition (Table I): 94% vs 62% on random scenarios, 92% vs 48% under doubled gusts, and 81% vs 56% out of distribution. It also settles faster (7.1 s vs 8.8 s on random scenarios), with less than half the residual sway (1.1° vs 2.5°). The cost is more aggressive actuation: GRPO saturates the force briefly at start-up and uses about 1.8× the control energy. The reward pays for speed and stillness, while u^2 is weighted lightly.

Where the RBF controller fails. With only the trolley actuated, the computed-traction law reaches the sway only through the inertia coupling of $M_0(q)$. Its quintic reference

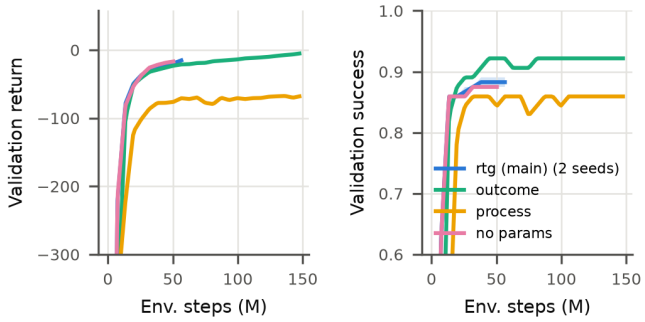


Fig. 2: GRPO validation return and success (64 held-out scenarios) vs. environment steps for the advantage variants and the no-parameter ablation.

is also fixed in advance, so the controller cannot exploit the sway. Online RBF adaptation barely helps: on our tuning set, success was 57% with adaptation and 63% with γ=0. Neural compensation of model error cannot create the missing actuation. The heatmap (Fig. 1) shows the RBF controller degrading quickly with longer cables (slower, more strongly coupled modes) and light payloads, where the hook dominates the dynamics. GRPO stays near 100% over most of the training box and degrades gracefully outside it, failing mainly beyond 2.6 m of cable, which was never seen in training.

Ablations (Fig. 2, random scenarios).

Compute. One GRPO iteration simulates 64×32 cranes for 600 steps (1.2M transitions) in about 12 s on an RTX 3050 Ti laptop GPU. GRPO passes 85% validation success within about 3 min. It is sample-hungry (tens of millions of steps), but cheap simulation makes this inexpensive in wall-clock time, and it needs no value network.

V. Conclusion

A critic-free, group-relative policy gradient, carried over from LLM reasoning to physical control, learns a robust anti-sway controller for the double-pendulum crane. Replaying the same scenario across a group gives a low-variance baseline for free, and a GPU-batched simulator makes the large number of rollouts practical. The learned policy outperforms a model-based RBF adaptive controller in positioning speed, residual sway and robustness to plant variation and gusts.

References

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